

LAKSHAY TUTEJA

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SUMMARY

Third-year CS undergraduate with expertise in **distributed data systems, full-stack development, and ML inference services**. From engineering hybrid **Spark/Kafka pipelines** at Samsung to leading a 14-person team to secure VIT's first software startup grant, I have a proven track record of delivering end-to-end solutions. Driven by a relentless **curiosity and a highly adaptable mindset**, I am eager to collaborate with like-minded builders to learn continuously and engineer the next generation of scalable systems.

EXPERIENCE

Samsung Prism, Research Intern

09/2025 – 02/2026

- Researched and implemented a hybrid Apache Spark application that unified Kafka-based streaming and batch processing under a **single Spark context**, eliminating the need to run separate pipelines for the same workload.
- Engineered the application to run as a **long-lived process**, dynamically accepting and serving concurrent streaming and batch requests until explicitly stopped, replacing the conventional one-job-per-run model.
- Worked under mentorship of a Samsung Senior Chief Engineer across a 6-month R&D cycle covering literature review, implementation in **Java/Scala with Spark 3.0+**, and end-to-end test cases.

LayOver, Founder & Technical Lead

12/2024 – 01/2026

- Founded and led a **14-person cross-functional team** spanning design, frontend, backend, and mobile development coordinating sprints, reviews, and deliverables across domains simultaneously.
- Architected the technical roadmap for a full-stack platform targeting integration with **30+ POS systems** across India, with planned payment gateway and third-party service layers.
- Secured **VIT's first institutional software startup grant** from the Ministry of Education / AICTE.

PROJECTS

Deepfake Image Detection Web App, Python · PyTorch · ResNet18 · FastAPI · React

- Fine-tuned a **ResNet18** model on a labelled dataset of real and AI-generated images to classify deepfakes, implementing the full training pipeline in PyTorch including data loading, augmentation, and evaluation.
- Exposed the model through a **FastAPI** inference endpoint that accepts image uploads and returns a confidence score with risk classification; connected to a React frontend.

Dear Diary Web App, Node.js · PostgreSQL (CockroachDB) · JWT · bcrypt · Gemini API · React

- Built a full-stack journaling app with a Node.js REST API and **PostgreSQL on CockroachDB**, implemented **JWT-based authentication and bcrypt password hashing** so credentials are unreadable even at the database level.
- **Integrated the Gemini API** to classify the emotional tone of each diary entry and surface it as an emoji on a React calendar view, connecting NLP output to a personalised UI.

SKILLS

Languages: JavaScript/TypeScript, Python, Java, Scala, SQL, HTML, CSS

Backend: FastAPI, Node.js, Apache Spark, Kafka, WebSockets, Docker

Databases: PostgreSQL, MongoDB, Redis, MySQL

AI / LLMs: PyTorch, OpenCV, MediaPipe, Gemini API, Google STT, AI/ML integrations

Frontend: React, ReactNative, Tailwind CSS, responsive design

EDUCATION

B.Tech Computer Science & Engineering, Vellore Institute of Technology, Vellore (VIT)

2023 – 2027

- CGPA: 8.53/10.00

Vellore, India

ACHIEVEMENTS

1st Place, HackBattle (VIT), VIT

- Built "Blinking Worlds" a 2D game controlled via real-time computer vision (OpenCV, MediaPipe) among 200+ participants.

Grant Awardee, VIT

- First software startup at VIT to receive exclusive institutional seed funding.